

Math

Ques:

Proof (1)

Given that,

$$a+ib=0$$

$$\text{or, } a=-ib.$$

By squaring we get,

$$a^2 - b^2$$

$$\text{or, } a^2 + b^2 = 0.$$

$$\therefore a=0 \text{ and } b=0.$$

Proof (2)

Given that,

$$a+ib = c+id.$$

$$\text{or, } (a-c) + i(b-d) = 0.$$

$$\therefore a-c=0 \text{ and } b-d=0 \text{ [As when } a+ib=0 \text{ then } a=0 \text{ \& } b=0]$$

$$\text{or, } a=c \text{ and } b=d.$$

Ques: Find the cube roots of unity.

Soln:

We know the cube roots of unity = $\sqrt[3]{1}$

Let,

$$\sqrt[3]{1} = x$$

$$\text{or, } x^3 - 1 = 0$$

$$\text{or, } (x-1)(x^2+x+1) = 0$$

$$\therefore (x-1) = 0 \quad \text{or, } x^2+x+1 = 0$$

When, $x-1=0$ then, $x=1$

When, $(x^2+x+1) = 0$ then,

$$x = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$

$$= \frac{-1 \pm i\sqrt{3}}{2} \quad [\because i^2 = -1]$$

$$= \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$$

$$\underline{\text{Ans}}: 1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$$

Ques: If $z = x + iy$ then find the locus ~~now~~ represented by $|z - 3| = 4$.

Soln:

Given that,

$$z = x + iy.$$

$$\text{Now, } |z - 3| = 4.$$

$$\text{or, } |x + iy - 3| = 4.$$

$$\text{or, } |(x - 3) + iy| = 4.$$

$$\text{or, } \sqrt{(x - 3)^2 + y^2} = 4.$$

$$\text{or, } (x - 3)^2 + y^2 = 4^2$$

$$\text{or, } (x - 3)^2 + (y - 0)^2 = 4^2.$$

which represents a circle whose centre is at $(3, 0)$ and radius 4.

Ans: circle

$$\text{Eqn. or, } x^2 + 6x + 9 + y^2 - 16 = 0.$$

or, $x^2 + y^2 + 6x - 7 = 0$, which is required eqn.

Ans:

De-Moivre's Theorem.

Ques:

If α and β be the roots of the eqn,
 $x^2 - 2x + 4 = 0$ then show that,

$$\alpha^n + \beta^n = 2^{n+1} \cos \frac{n\pi}{3}.$$

Soln:

Given the eqn,

$$x^2 - 2x + 4 = 0.$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 4}}{2 \cdot 1}$$

$$= \frac{2 \pm \sqrt{4 - 16}}{2}$$

$$= \frac{2 \pm \sqrt{-12}}{2}$$

$$= \frac{2 \pm \sqrt{-3 \times 4}}{2}$$

$$= \frac{2 \pm 2i\sqrt{3}}{2}$$

$$= 1 \pm i\sqrt{3}$$

$$= 1 + i\sqrt{3}, 1 - i\sqrt{3}$$

According to Question,

$$\alpha = 1 + i\sqrt{3} \text{ and } \beta = 1 - i\sqrt{3}.$$

Now,

$$\alpha^n + \beta^n = (1 + i\sqrt{3})^n + (1 - i\sqrt{3})^n$$

$$= (r \cos \theta + i r \sin \theta)^n + (r \cos \theta - i r \sin \theta)^n$$

$$= r^n (\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta)$$

$$= r^n \cdot 2 \cos n\theta$$

$$= 2^n \cdot 2 \cos n \frac{\pi}{3}$$

$$= 2^{n+1} \cos \frac{n\pi}{3}$$

Let,

$$1 = r \cos \theta$$

$$2\sqrt{3} = r \sin \theta$$

$$\therefore r = \sqrt{1+3}$$

$$= 2.$$

And,

$$\theta = \tan^{-1} \sqrt{3}.$$

$$= \frac{\pi}{3}.$$

Ques: Obtain the quadratic equation having the roots of n^{th} power of the roots of $x^2 - 2x \cos \theta + 1 = 0$

Soln: Given the eqn,

$$x^2 - 2x \cos \theta + 1 = 0$$

$$\therefore x = \frac{-(-2\cos\theta) \pm \sqrt{(-2\cos\theta)^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$= \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2}$$

$$= \frac{2\cos\theta \pm \sqrt{-4(1-\cos^2\theta)}}{2}$$

$$= \frac{2\cos\theta \pm i 2\sin\theta}{2} \quad [\because i^2 = -1]$$

$$= \cos\theta \pm i\sin\theta$$

According to ques: the roots of the required equation will be,

$$\alpha^n = (\cos\theta + i\sin\theta)^n$$

$$= \cos n\theta + i\sin n\theta$$

and,

$$\beta^n = (\cos\theta - i\sin\theta)^n$$

$$= \cos n\theta - i\sin n\theta$$

Hence the required eqn,

$$\alpha^n - (\alpha^n + \beta^n)\alpha + \alpha^n\beta^n = 0$$

$$\text{or, } x^n - (\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta)x - (\cos n\theta + i \sin n\theta)(\cos n\theta - i \sin n\theta) = 0$$

$$\text{or, } x^n - (2\cos n\theta)x + (\cos^2 n\theta + \sin^2 n\theta) = 0$$

$$\text{or, } x^n - 2x\cos n\theta + 1 = 0 \quad [\because i^2 = -1]$$

[Ans].

Ques: Show that,

$$(a+ib)^{\frac{m}{n}} + (a-ib)^{\frac{m}{n}} = 2(a^2+b^2)^{\frac{m}{2n}} \cos\left(\frac{m}{n} \tan^{-1} \frac{b}{a}\right)$$

Soln:

Given the eqn,

$$(a+ib)^{\frac{m}{n}} + (a-ib)^{\frac{m}{n}} = 2(a^2+b^2)^{\frac{m}{2n}} \cos\left(\frac{m}{n} \tan^{-1} \frac{b}{a}\right)$$

L.S.,

$$(a+ib)^{\frac{m}{n}} + (a-ib)^{\frac{m}{n}}$$

$$= (r\cos\theta + ir\sin\theta)^{\frac{m}{n}} + (r\cos\theta - ir\sin\theta)^{\frac{m}{n}}$$

$$= r^{\frac{m}{n}} \left(\cos^{\frac{m}{n}}\theta + i \sin^{\frac{m}{n}}\theta + \cos^{\frac{m}{n}}\theta - i \sin^{\frac{m}{n}}\theta \right)$$

$$= \sqrt{a^2+b^2}^{\frac{m}{n}} \left(2\cos^{\frac{m}{n}} \tan^{-1} \frac{b}{a} \right)$$

Let,

$$a = r\cos\theta$$

$$b = r\sin\theta$$

$$r = \sqrt{a^2+b^2}$$

$$\theta = \tan^{-1} \frac{b}{a}$$

$$\therefore 2(a^2+b^2)^{\frac{1}{2} \frac{m}{n}} \left(\cos \frac{m}{n} \tan^{-1} \frac{b}{a} \right).$$

$$\therefore 2(a^2+b^2)^{\frac{m}{2n}} \cos \left(\frac{m}{n} \tan^{-1} \frac{b}{a} \right).$$

Ans

Ques:

Solve the equation,

$$x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0.$$

Using De-Moivre's theorem,

Soln:

Given the eqn:

$$x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$$

$$\text{or, } (x^6 + x^5 + x^4 + x^3 + x^2 + x + 1)(x-1) = 0$$

$$\text{or, } (x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x) - (x^6 + x^5 + x^4 + x^3 + x^2 + x + 1) = 0.$$

$$\text{or, } x^7 - 1 = 0.$$

$$\text{or, } x = (1)^{\frac{1}{7}}$$

$$= (1 - i \cdot 0)^{\frac{1}{7}}$$

$$= (r \cos \theta + i r \sin \theta)^{\frac{1}{7}}$$

$$= r^{\frac{1}{7}} (\cos \theta + i \sin \theta)^{\frac{1}{7}}$$

Let,

$1 = r \cos \theta$

and, $0 = r \sin \theta$

$r = \sqrt{1^2 + 0^2}$

and, $\theta = \tan^{-1} 0 = 0$

$$= 1^{\frac{1}{7}} (\cos 0 + i \sin 0)^{\frac{1}{7}}$$

$$= (\cos 0 + i \sin 0)^{\frac{1}{7}}$$

$$= \left\{ \cos(2n\pi + 0) + i \sin(2n\pi + 0) \right\}^{\frac{1}{7}},$$

$$[n=0, 1, 2, \dots]$$

$$(i) \quad \cos \frac{2n\pi}{7} + i \sin \frac{2n\pi}{7}$$

$$(ii) \quad \dots$$

Now putting $n=0, 1, 2, 3, 4, 5, 6$ we get,

$$n=0 \quad \cos 0 + i \sin 0 = 1, \quad n=0, \text{ not excepted}$$

$$= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}, \quad n=1,$$

$$= \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7}, \quad n=2,$$

$$= \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7}, \quad n=3,$$

$$= \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}, \quad n=4,$$

$$= \cos \frac{10\pi}{7} + i \sin \frac{10\pi}{7}, \quad n=5,$$

$$= \cos \frac{12\pi}{7} + i \sin \frac{12\pi}{7}, \quad n=6,$$

Ans

Ques: Using De Moivre's theorem, find the value of $(\sqrt{3} + i \cdot 1)^{1/5}$.

Soln:

$$\text{Let, } z = (\sqrt{3} + i \cdot 1)^{1/5}$$

$$\sqrt{3} = r \cos \theta \quad \dots \quad (i)$$

$$1 = r \sin \theta \quad \dots \quad (ii)$$

Squaring, (i) & (ii),

$$3 = r^2 \cos^2 \theta \quad \dots \quad (iii)$$

$$1 = r^2 \sin^2 \theta \quad \dots \quad (iv)$$

Adding (iii) & (iv),

$$3 + 1 = r^2 \cos^2 \theta + r^2 \sin^2 \theta$$

$$\Rightarrow 4 = r^2$$

$$\therefore r = 2 \quad \dots \quad (v)$$

$$(iii) \div (i),$$

$$\frac{1}{\sqrt{3}} = \frac{r \sin \theta}{r \cos \theta}$$

$$\text{or, } \frac{1}{\sqrt{3}} = \tan \theta$$

$$\text{or, } \tan \frac{\pi}{6} = \tan \theta$$

$$\therefore \theta = \frac{\pi}{6} \quad \dots \quad (vi)$$

Given,

$$(\sqrt{3} + i \cdot 1)^{\frac{1}{5}}$$

$$= (r \cos \theta + i \cdot r \sin \theta)^{\frac{1}{5}}$$

$$= \{ r (\cos \theta + i \sin \theta) \}^{\frac{1}{5}}$$

$$= \{ 2^{\frac{1}{5}} \{ \cos(2n\pi + \frac{\pi}{6}) + i \sin(2n\pi + \frac{\pi}{6}) \} \}^{\frac{1}{5}}$$

$$= \{ 2^{\frac{1}{5}} \{ \cos(\frac{12n\pi + \pi}{6}) + i \sin(\frac{12n\pi + \pi}{6}) \} \}^{\frac{1}{5}}$$

$$= \{ 2^{\frac{1}{5}} \{ \cos(\frac{12n\pi + \pi}{30}) + i \sin(\frac{12n\pi + \pi}{30}) \} \}$$

$$[\cos \theta + i \sin \theta]^n$$

$$\cos n\theta + i \sin n\theta$$

Now, putting $n = 0, 1, 2, 3, 4$

$$= \{ 2^{\frac{1}{5}} \{ \cos(\frac{\pi}{30}) + i \sin(\frac{\pi}{30}) \} \}, n=0.$$

$$= \{ 2^{\frac{1}{5}} \{ \cos(\frac{13\pi}{30}) + i \sin(\frac{13\pi}{30}) \} \}, n=1.$$

$$= \{ 2^{\frac{1}{5}} \{ \cos(\frac{25\pi}{30}) + i \sin(\frac{25\pi}{30}) \} \}, n=2.$$

$$= \{ 2^{\frac{1}{5}} \{ \cos(\frac{37\pi}{30}) + i \sin(\frac{37\pi}{30}) \} \}, n=3.$$

$$= \{ 2^{\frac{1}{5}} \{ \cos(\frac{49\pi}{30}) + i \sin(\frac{49\pi}{30}) \} \}, n=4.$$

Ans

Ques: Using De Moivre's theorem, find the value of $(1+i)^{1/5}$.

Soln: Let, $z = (1+i)^{1/5}$

$$1 = r \cos \theta \quad \dots \dots (i)$$

$$1 = r \sin \theta \quad \dots \dots (ii)$$

Squaring, (i) & (ii),

$$1 = r^2 \cos^2 \theta \quad \dots \dots (iii)$$

$$1 = r^2 \sin^2 \theta \quad \dots \dots (iv)$$

Adding, (iii) & (iv)

$$1+1 = r^2 \cos^2 \theta + r^2 \sin^2 \theta$$

$$\text{or, } 2 = r^2 (\cos^2 \theta + \sin^2 \theta)$$

$$\text{or, } r = \sqrt{2} \quad \dots \dots (v)$$

Now

$$(ii) \div (i),$$

$$\frac{r \sin \theta}{r \cos \theta} = 1$$

$$\text{or, } \tan \theta = 1$$

$$\text{or, } \theta = \frac{\pi}{4} \quad \dots \dots (vi)$$

Given,

$$(1 + i)^{1/5}$$

$$= (r \cos \theta + i \cdot r \sin \theta)^{1/5}$$

$$= \{ r (\cos \theta + i \sin \theta) \}^{1/5}$$

$$= \{ \sqrt{2} \}^{1/5} \{ \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \}^{1/5}$$

$$= (\sqrt{2})^{1/5} \{ \cos (\frac{2\pi n}{4} + \frac{\pi}{4}) + i \sin (\frac{2\pi n}{4} + \frac{\pi}{4}) \}^{1/5}$$

$$= (\sqrt{2})^{1/5} \{ \cos (\frac{8\pi n + \pi}{4}) + i \sin (\frac{8\pi n + \pi}{4}) \}^{1/5}$$

$$= (\sqrt{2})^{1/5} \{ \cos (\frac{8\pi n + \pi}{20}) + i \sin (\frac{8\pi n + \pi}{20}) \}$$

$$[\cos \theta + i \sin \theta]^n = \cos n\theta + i \sin n\theta$$

Now, putting $n = 0, 1, 2, 3, 4$.

$$x = \{ (\sqrt{2})^{1/5} \cos$$

$$n = [(\sqrt{2})^{1/5} \{ \cos (\frac{\pi}{20}) + i \sin (\frac{\pi}{20}) \}], n=0.$$

$$= [(\sqrt{2})^{1/5} \{ \cos (\frac{9\pi}{20}) + i \sin (\frac{9\pi}{20}) \}], n=1$$

$$= [(\sqrt{2})^{1/5} \{ \cos (\frac{17\pi}{20}) + i \sin (\frac{17\pi}{20}) \}], n=2$$

$$= [(\sqrt{2})^{1/5} \{ \cos (\frac{25\pi}{20}) + i \sin (\frac{25\pi}{20}) \}], n=3$$

$$= [(\sqrt{2})^{1/5} \{ \cos (\frac{33\pi}{20}) + i \sin (\frac{33\pi}{20}) \}], n=4$$

Ques: If $\alpha_n = \cos \frac{\pi}{2^n} + i \sin \frac{\pi}{2^n}$ then show that, $\alpha_1, \alpha_2, \alpha_3, \dots, \infty = -1$.

Soln: Given that,

$$\alpha_n = \cos \frac{\pi}{2^n} + i \sin \frac{\pi}{2^n} \dots \text{--- (1)}$$

Now,

putting $n = 1, 2, 3, \dots, \infty$ we

have,

$$\alpha_1 = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$\alpha_2 = \cos \frac{\pi}{2^2} + i \sin \frac{\pi}{2^2}$$

$$\alpha_3 = \cos \frac{\pi}{2^3} + i \sin \frac{\pi}{2^3}$$

Now, L.S. $\alpha_1, \alpha_2, \alpha_3, \dots, \infty$

$$= \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \left(\cos \frac{\pi}{2^2} + i \sin \frac{\pi}{2^2} \right)$$

$$\cdot \left(\cos \frac{\pi}{2^3} + i \sin \frac{\pi}{2^3} \right) \dots \infty$$

$$= \cos \left(\frac{\pi}{2} + \frac{\pi}{2^2} + \frac{\pi}{2^3} + \dots \infty \right) + i \sin \left(\frac{\pi}{2} + \frac{\pi}{2^2} + \frac{\pi}{2^3} + \dots \infty \right)$$

$$= \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right)$$

[As $1 + ar + ar^2 + ar^3 + \dots$

$$= \frac{a}{1-r}, r < 1]$$

$$= \cos\left(\frac{\pi \times 2}{2}\right) + i \sin\left(\frac{\pi \times 2}{2}\right)$$

$$= \cos \pi + i \sin \pi$$

$$= -1$$

$$= R.S.$$

Hence

Proved

Ques: If $(1+i\frac{x}{a})(1+i\frac{x}{b})(1+i\frac{x}{c})\dots\dots = A+iB$.

Then, prove that

(i) $(1+\frac{x^2}{a^2})(1+\frac{x^2}{b^2})(1+\frac{x^2}{c^2})\dots\dots = A^2+B^2$

Soln:

Given, $(1+i\frac{x}{a})(1+i\frac{x}{b})(1+i\frac{x}{c})\dots\dots = A+iB$
..... (i)

Let, $1 = r \cos \alpha, \frac{x}{a} = r \sin \alpha$

Again let, $1 = r \cos \beta, \frac{x}{b} = r \sin \beta$

Again let, $1 = r \cos \gamma, \frac{x}{c} = r \sin \gamma$

So, $\frac{\frac{x}{a}}{1} = \frac{r \sin \alpha}{r \cos \alpha}$

$\therefore \frac{x}{a} = \tan \alpha$

$\therefore \alpha = \tan^{-1}\left(\frac{x}{a}\right)$

So, $\frac{\frac{x}{b}}{1} = \frac{r \sin \beta}{r \cos \beta}$

$\therefore \tan \beta = \frac{x}{b}$

$\therefore \beta = \tan^{-1}\left(\frac{x}{b}\right)$

So,

$$\frac{\frac{a}{c}}{1} = \frac{r \sin \gamma}{r \cos \gamma}$$

$$\therefore \gamma = \tan^{-1}\left(\frac{a}{c}\right)$$

From (i), we get,

$$\text{Given, } (1+i\frac{a}{a})(1+i\frac{a}{b})(1+i\frac{a}{c})\dots = A+ib$$

$$\Rightarrow (1+i \tan \alpha)(1+i \tan \beta)(1+i \tan \gamma)\dots = A+ib$$

$$\Rightarrow \left(1+i \frac{\sin \alpha}{\cos \alpha}\right) \left(1+i \frac{\sin \beta}{\cos \beta}\right) \left(1+i \frac{\sin \gamma}{\cos \gamma}\right)\dots = A+ib$$

$$\Rightarrow \left(\frac{\cos \alpha + i \sin \alpha}{\cos \alpha}\right) \left(\frac{\cos \beta + i \sin \beta}{\cos \beta}\right) \left(\frac{\cos \gamma + i \sin \gamma}{\cos \gamma}\right)\dots = A+ib$$

$$\Rightarrow \left(\frac{1}{\cos \alpha}\right) \left(\frac{1}{\cos \beta}\right) \left(\frac{1}{\cos \gamma}\right)\dots (\cos \alpha + i \sin \alpha) (\cos \beta + i \sin \beta) (\cos \gamma + i \sin \gamma) \dots = A+ib$$

$$\Rightarrow (\sec \alpha)(\sec \beta)(\sec \gamma)\dots (\cos \alpha + i \sin \alpha) (\cos \beta + i \sin \beta) (\cos \gamma + i \sin \gamma)\dots = A+ib$$

$$\Rightarrow (\sec \alpha)(\sec \beta)(\sec \gamma)\dots [\cos (\alpha + \beta + \gamma + \dots) + i \sin (\alpha + \beta + \gamma + \dots)] = A+ib$$

Equating the real and imaginary part on both sides, we get,

$$\begin{aligned} (\sec \alpha)(\sec \beta)(\sec \gamma) \dots \dots [\cos(\alpha + \beta + \gamma + \dots)] &= A \dots \dots \text{(ii)} \\ (\sec \alpha)(\sec \beta)(\sec \gamma) \dots \dots [\sin(\alpha + \beta + \gamma + \dots)] &= B \dots \dots \text{(iii)} \end{aligned}$$

Squaring (ii) and (iii), we get,

$$\begin{aligned} (\sec^2 \alpha)(\sec^2 \beta)(\sec^2 \gamma) \dots \dots [\cos^2(\alpha + \beta + \gamma + \dots)] &= A^2 \dots \dots \text{(iv)} \\ (\sec^2 \alpha)(\sec^2 \beta)(\sec^2 \gamma) \dots \dots [\sin^2(\alpha + \beta + \gamma + \dots)] &= B^2 \dots \dots \text{(v)} \end{aligned}$$

Adding (iv) & (v)

$$\begin{aligned} &(\sec^2 \alpha)(\sec^2 \beta)(\sec^2 \gamma) \dots \dots \cos^2(\alpha + \beta + \gamma + \dots) \\ + &(\sec^2 \alpha \sec^2 \beta \sec^2 \gamma) \sin^2(\alpha + \beta + \gamma + \dots) = A^2 + B^2 \\ \Rightarrow &(\sec^2 \alpha \sec^2 \beta \sec^2 \gamma) \{ \cos^2(\alpha + \beta + \gamma + \dots) + \sin^2(\alpha + \beta + \gamma + \dots) \} \\ &= A^2 + B^2 \end{aligned}$$

$$\Rightarrow (\sec^2 \alpha \sec^2 \beta \sec^2 \gamma) \cdot 1 = A^2 + B^2$$

$$\Rightarrow (1 + \tan^2 \alpha)(1 + \tan^2 \beta)(1 + \tan^2 \gamma) \dots \dots = A^2 + B^2$$

$$\therefore \left(1 + \frac{a^2}{b^2}\right) \left(1 + \frac{b^2}{c^2}\right) \left(1 + \frac{c^2}{a^2}\right) \dots \dots = A^2 + B^2$$

Proved

Ques: If $2\cos\alpha = a + \frac{1}{a}$, $2\cos\beta = b + \frac{1}{b}$, $2\cos\gamma = c + \frac{1}{c}$
and so on, then,

Show that,

$$2\cos(\alpha + \beta + \gamma + \dots) = abc \dots + \left(\frac{1}{abc \dots}\right)$$

Soln: Given that,

$$2\cos\alpha = a + \frac{1}{a}$$

$$2\cos\beta = b + \frac{1}{b}$$

$$2\cos\gamma = c + \frac{1}{c}$$

Complex Mapping

Ques: Determine the image of the point P, $z = 3 + j2$ on the w-plane under the transformation $w = 3z + 2 - j$.

Solⁿ:

Given that,

$$w = 3z + 2 - j$$

$$\begin{aligned} \text{or, } u + jv &= 3(x + jy) + 2 - j \\ &= 3x + j3y + 2 - j \\ &= 3x + 2 + j(3y - 1). \end{aligned}$$

Now equating real and imaginary parts from both sides

We have,

$$u = 3x + 2 \dots \text{--- (i)}$$

$$\text{and } v = 3y - 1 \text{ --- (ii)}$$

Again, Given the point P,
 $z = 3 + j2$.

Here,

$$x = 3 \text{ and } y = 2.$$

$$\therefore P(3, 2).$$

Now, putting the values of x & y in

① & ② we get,

$$u = 3 \cdot 3 + 2 \cdot 2 = 11$$

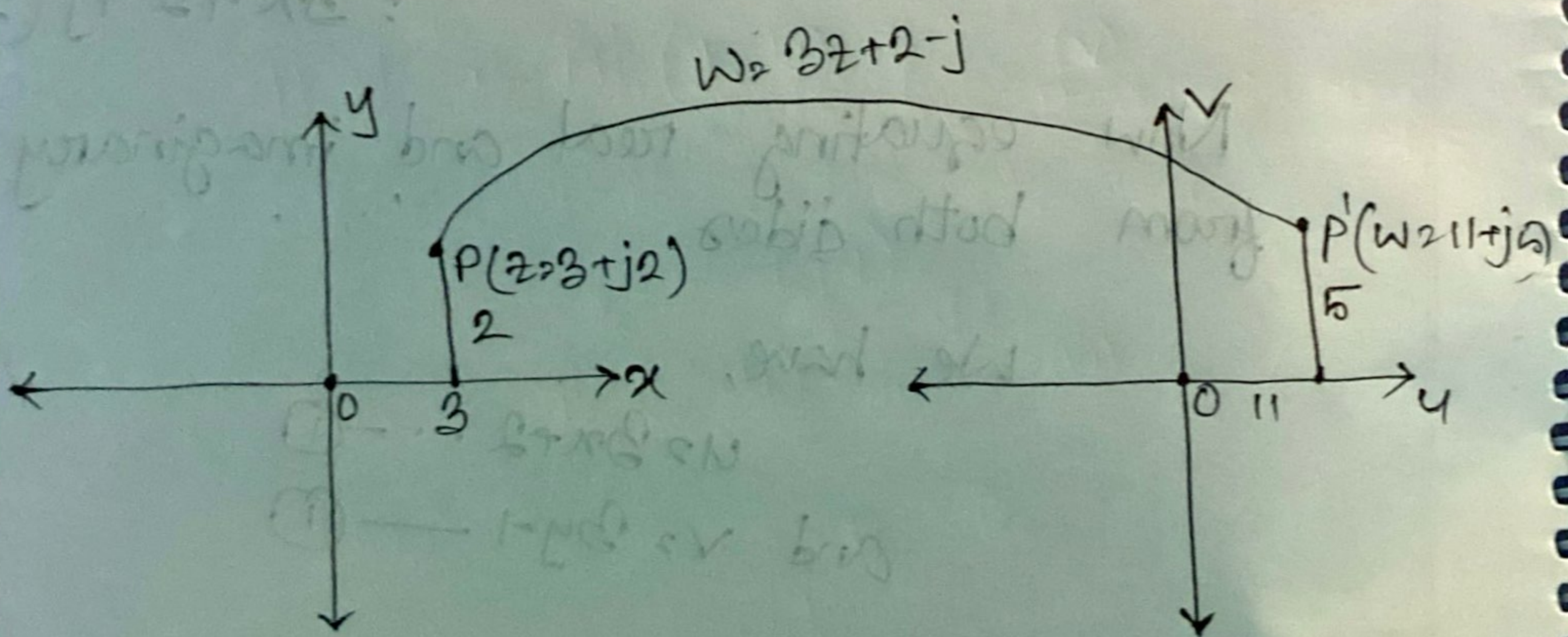
$$\text{and, } v = 3 \cdot 2 - 1 \cdot 5 = 5$$

Hence,

the image of P is,

$$P'(11, 5), \quad P' = W = u + jv \\ = 11 + j5$$

Ans



z plane $\rightarrow$ w plane

Ques: Map the point A ($z = -2+j$) and B ($z = 3+j4$) on the w-plane under the transformation $w = j2z+3$ and illustrate the transformation on a diagram.

Soln:

We have,

$$w = f(z) = j2z+3.$$

$$u+jv = f(z) = j2z+3. \quad [w = u+jv]$$

$$u+jv = j2(x+jy)+3. \quad [z = x+jy]$$

$$u+jv = j2x + 2j^2y + 3.$$

$$u+jv = j2x - 2y + 3 \quad [j^2 = -1].$$

$$u+jv = (3-2y) + j2x. \quad \dots (i)$$

Equating the coefficient of real and imaginary part, we get,

$$\therefore u = 3-2y. \quad \dots (ii)$$

$$\therefore v = 2x. \quad \dots (iii)$$

Given, A ($z = -2+j \cdot 1$)

That is, A (-2, 1) $\dots (iv)$

Here,

$$x = -2, y = 1$$

Putting the value of x and y in (ii) and (iii),

$$u = 3 - 2y$$

$$u = 3 - 2 \cdot 1$$

$$u = 1$$

$$v = 2x$$

$$v = 2 \cdot (-2)$$

$$v = -4$$

$$\therefore w = u + jv$$

$$= 1 - j4$$

The image of A is $A' (w = 1 - j4)$.

That is $A' (1 - j4) \dots \dots \dots \underline{\underline{(v)}}$

Again,

$$B (z = 3 + j4)$$

That, $B (3, 4) \dots \dots \dots \underline{\underline{(vi)}}$

Here, $x = 3, y = 4$

Putting the value of x & y in (ii) and (iii),

$$u = 3 - 2y$$

$$u = 3 - 2 \cdot 4$$

$$\therefore u = -5$$

$$v = 2x$$

$$v = 2 \cdot 3$$

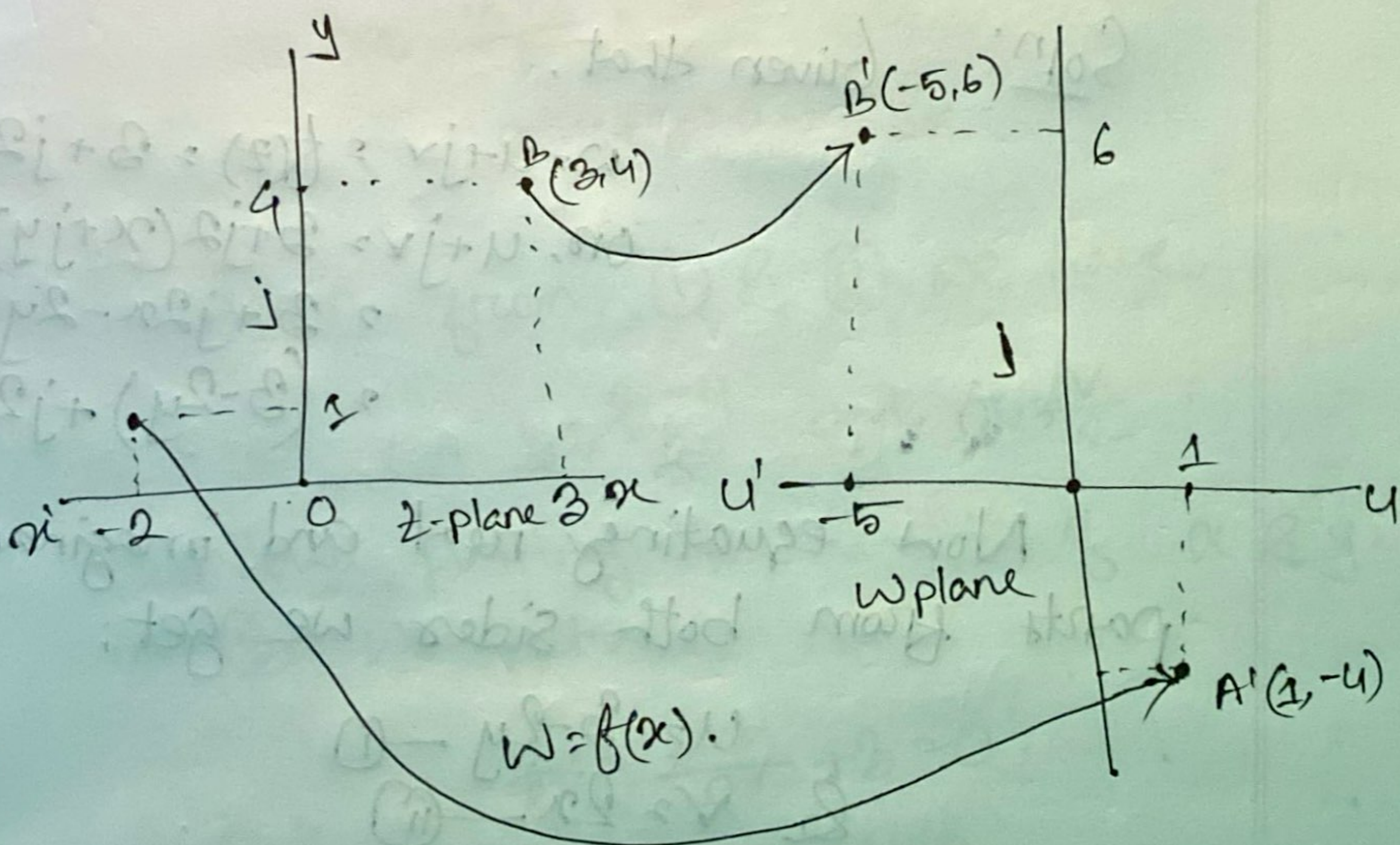
$$v = 6$$

$$\therefore w = u + jv = -5 + j \cdot 6$$

The image of B is $B' (w = u + jv)$

$$= -5 + j \cdot 6$$

That is $B' (-5, 6) \dots (v)$



Ques: To map the straight line joining $A(-2+j)$ and $B(3+j6)$ in the z -plane on to the w -plane, when $w = u+jv$ $f(z) = 3+j2z$.

Soln: Given that,

$$w = u+jv = f(z) = 3+j2z.$$

$$\text{or, } u+jv = 3+j2(x+jy).$$

$$= 3+j2x-2y [\because j^2 = -1]$$

$$= (3-2y) + j2x.$$

Now equating real and imaginary parts from both sides we get,

$$u = 3-2y \quad \text{--- (i)}$$

$$\& \quad v = 2x \quad \text{--- (ii)}$$

Again the given points are

$$A(-2+j)$$

$$\therefore A(-2, 1).$$

Also,

$$B(3+j6)$$

$$\therefore B(3, 6).$$

Now, the line joining the points A & B is given by:

$$\frac{x+2}{-2-3} = \frac{y-1}{1-6}$$

$$\text{or, } \frac{x+2}{-5} = \frac{y-1}{-5}$$

$$\text{or, } x-y+3=0 \quad \text{--- (3)}$$

Again, from (1) & (2) we have,

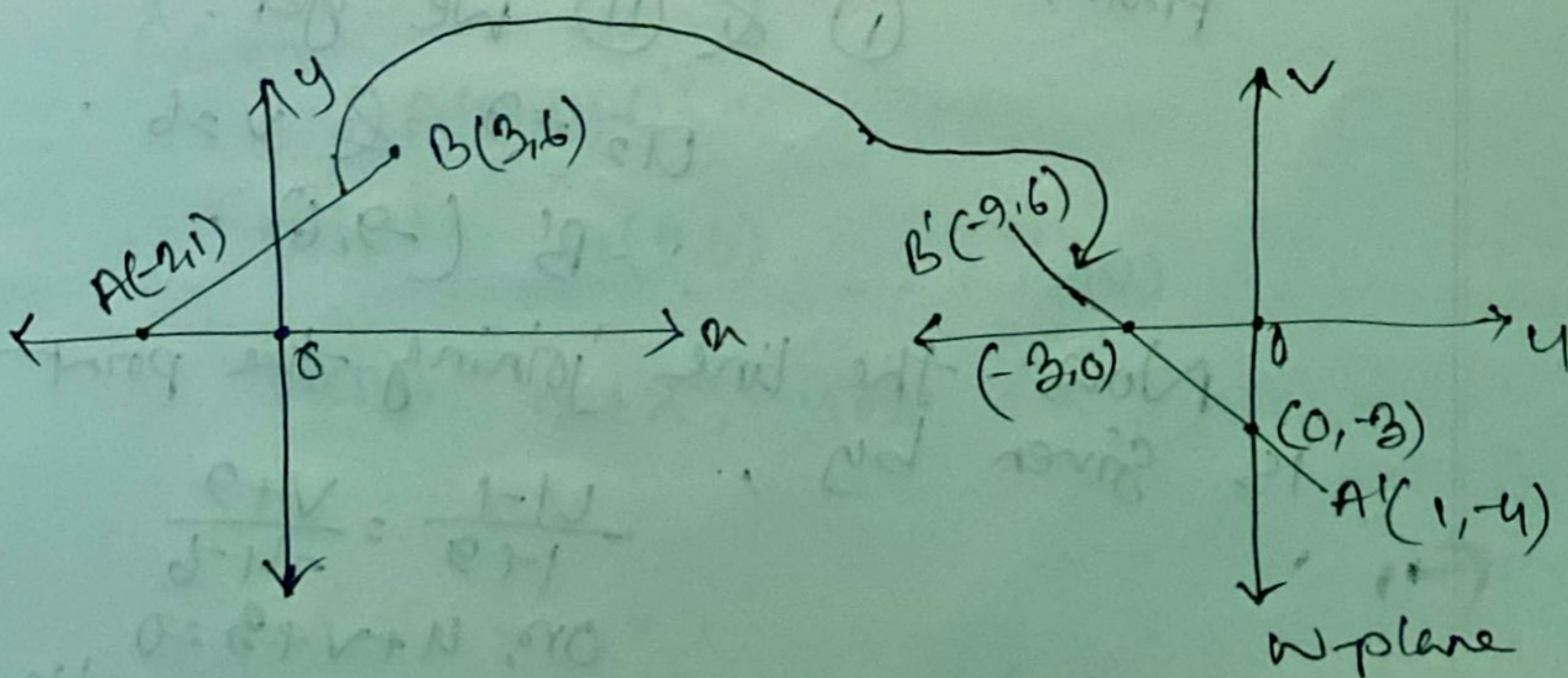
$$y = \frac{3-u}{2} \quad \text{and, } x = \frac{v}{2}$$

Now, putting the values of x & y in (3) we get,

$$\frac{v}{2} - \frac{3-u}{2} + 3 = 0$$

$$\text{or, } v-3+u+6=0$$

$$\text{or, } u+v+3=0, \text{ which is required}$$



$$u + v = -3,$$

$$\frac{u}{-3} + \frac{v}{-3} = 1$$

$$\frac{(-3, 0)}{A'}, \frac{(0, -3)}{B'}$$

Justification:

Given the points are $A(-2, 1)$

$$\therefore x_1 = -2, \text{ \& } y_1 = 1$$

From (i) & (ii) we get, $u = 1$ & $v = -4$

$$\therefore A'(1, -4)$$

Also,

$$B(3, 6)$$

$$\therefore x_2 = 3, \text{ \& } y_2 = 6$$

From,

(i) & (ii) we get,

$$u = -9 \text{ \& } v = 6$$

$$\therefore B'(-9, 6)$$

Now, the line joining the point A' & B' is given by,

$$\frac{u+1}{1+9} = \frac{v+9}{-4-6}$$

$$\text{Or, } u + v + 3 = 0$$

Hence justified

Ques:

If $w = z^2$, find the path traced out by w as z move along the straight line joining $A(2+j.0)$ and $B(0+2j)$.

Soln:

We have, $w = f(z) = z^2$

$$u + jv = f(z) = (x + jy)^2 \quad [w = u + jv; \quad z = x + jy]$$

$$u + jv = x^2 + j2xy + j^2 y^2$$

$$u + jv = x^2 + j2xy - y^2 \quad [\because j^2 = -1]$$

$$u + jv = x^2 - y^2 + j2xy \dots \dots (i)$$

Equating the coefficient of real and imaginary part, we get

$$\therefore u = x^2 - y^2 \dots \dots (ii)$$

$$\therefore v = 2xy \dots \dots (iii)$$

Given,

$$A(2+j.0)$$

$$\text{That is, } A(2,0) \dots \dots (iv)$$

$$\text{Here, } x = 2, y = 0.$$

putting the values of x & y in (ii) and (iii),

$$\begin{aligned}u &= x^2 + y^2 \\u &= 0^2 + 2^2 \\u &= 4\end{aligned}$$

$$\begin{aligned}v &= 2xy \\v &= 2 \cdot 0 \cdot 2 \\v &= 0\end{aligned}$$

$$u = 4$$

$$\therefore W = u + jv \\= 4 + j \cdot 0$$

That is $B'(-4, 0) \dots \dots (vii)$

We have, $A(2, 0)$ and $B(0, 2)$.

$$A: x_1 = 2, y_1 = 0$$

$$B: x_2 = 0, y_2 = 2$$

The equation of the line AB is,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 0}{0 - 2} = \frac{x - 2}{2 - 0}$$

$$\Rightarrow \frac{y}{-2} = \frac{x - 2}{2}$$

$$\Rightarrow 2y = -2(x-2)$$

$$\Rightarrow y = -(x-2)$$

$$\therefore y = 2-x \dots \dots (viii)$$

putting the value of y in (ii),

$$\therefore u = x^2 - y^2$$

$$\Rightarrow u = x^2 - (2-x)^2$$

$$\Rightarrow u = x^2 - (4 - 4x + x^2)$$

$$\Rightarrow u = x^2 - 4 + 4x - x^2$$

$$\Rightarrow u = 4x - 4$$

$$\Rightarrow x = \frac{u+4}{4} \dots \dots (ix)$$

putting the value of y in (iii),

$$v = 2xy$$

$$v = 2x(2-x)$$

$$\therefore v = 4x - 2x^2 \dots \dots (x)$$

Now putting the values of x in eqn (x)

$$\therefore v = 4x - 2x^2$$

$$\Rightarrow v = 4\left(\frac{u+4}{4}\right) - 2\left(\frac{u+4}{4}\right)^2$$

$$\Rightarrow \sqrt{2} \cdot u + 4 - 2 \left(\frac{u^2 + 8u + 16}{16} \right)$$

$$\Rightarrow \sqrt{2} \cdot u + 4 - \frac{1}{8} (u^2 + 8u + 16)$$

$$\Rightarrow \sqrt{2} \cdot u + 4 - \frac{1}{8} u^2 - u - 2$$

$$\Rightarrow \sqrt{2} \cdot 2 - \frac{1}{8} u^2$$

$$\Rightarrow \sqrt{2} \cdot \frac{16 - u^2}{8}$$

$$\Rightarrow \sqrt{2} - \frac{1}{8} (u^2 - 16)$$

$$\Rightarrow 8\sqrt{2} - u^2 + 16$$

$$\Rightarrow -u^2 = 8\sqrt{2} - 16$$

$$\Rightarrow u^2 = -8(\sqrt{2} - 2)$$

$$\Rightarrow u^2 = -4 \cdot 2(\sqrt{2} - 2) \dots (xi)$$

The equation (xi) represents an equation of a parabola,

Let, $u = v$ and $\sqrt{2} = v - 2 \dots (xii)$

From (xii)

when,

$u = 0$ then $u = 0$

and,

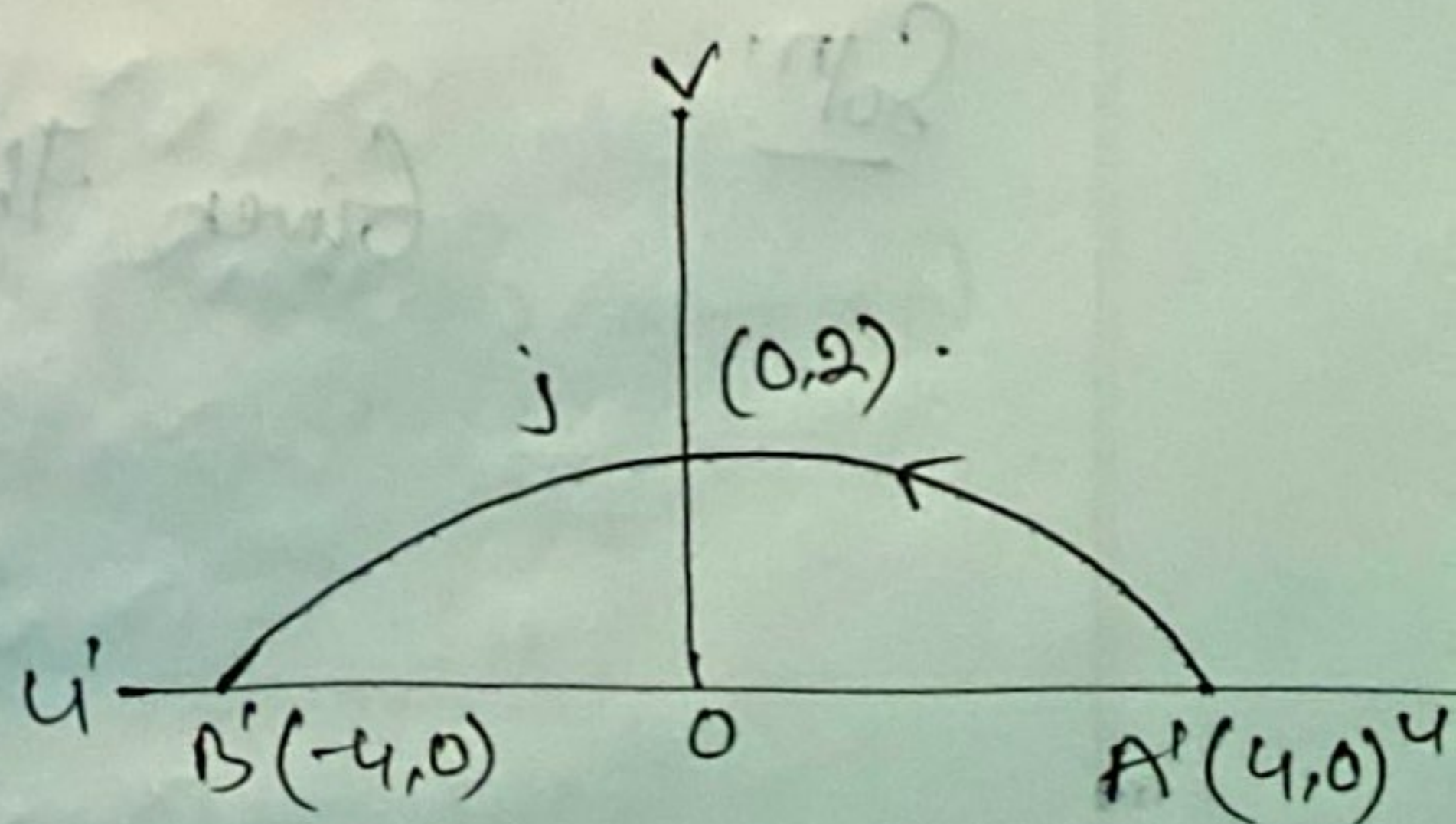
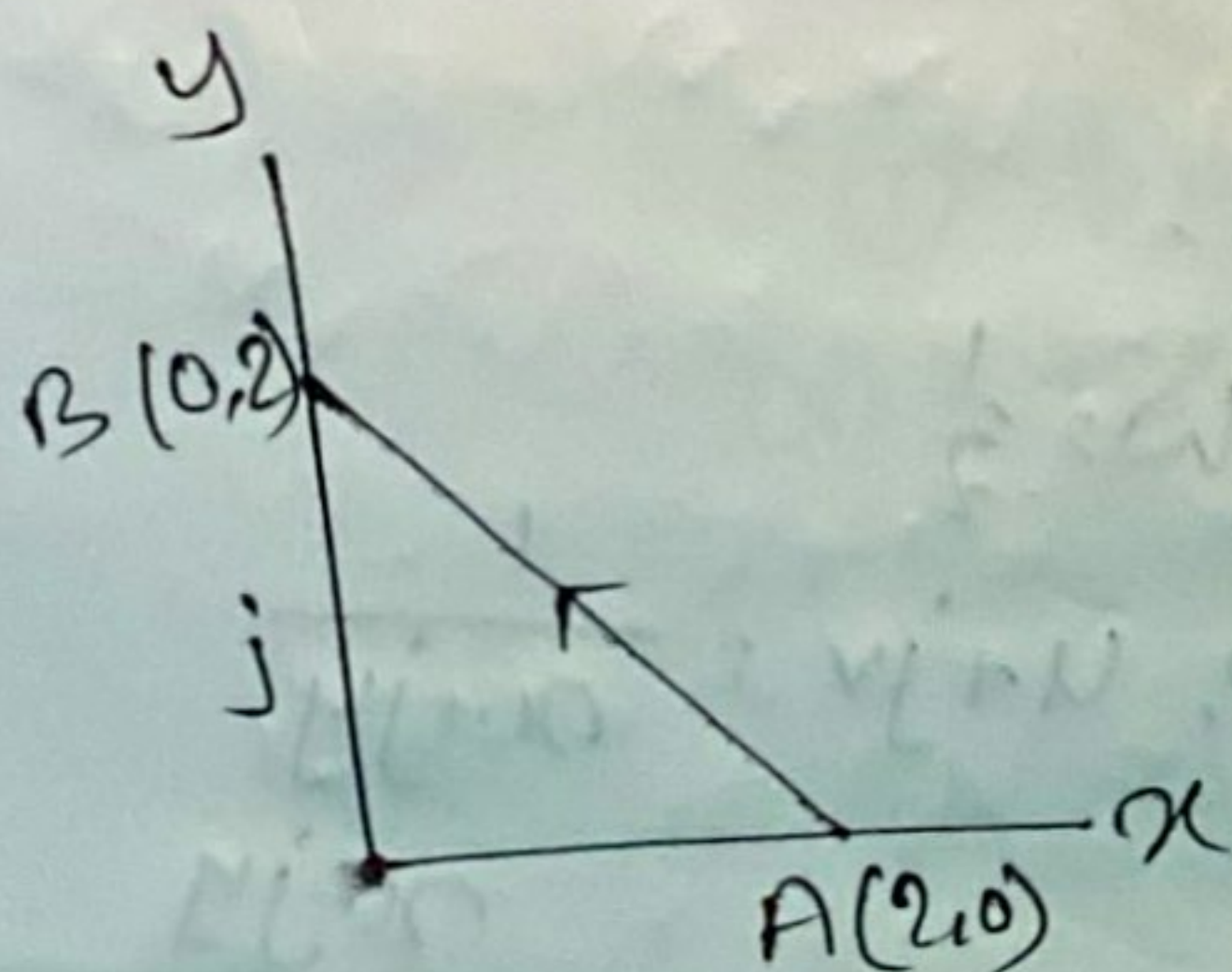
$\sqrt{2} = 0$ then

$$\Rightarrow 0 = \sqrt{2} - 2$$

$$\Rightarrow v = 2.$$

$$\therefore \text{vertex } (u, v) = (0, 2).$$

$$\begin{aligned} \text{And latus rectum } A'B' &= 4a \\ &= 4 \cdot 2 \\ &= 8. \end{aligned}$$



Ques:

A circle in the z -plane has its center at $z=3$ and a radius of 2 unit.

Determine its image in the w -plane when transformation by $w=\frac{1}{z}$.

Soln:

Given that,

$$w = \frac{1}{z}$$

$$\text{or, } u+jv = \frac{1}{x+jy}$$

$$= \frac{x-jy}{x^2+y^2} \quad [j^2 = -1]$$

$$= \frac{x}{x^2+y^2} + j \left(\frac{-y}{x^2+y^2} \right)$$

Now equating real and imaginary parts from both sides we get,

$$u = \frac{x}{x^2+y^2} \quad \text{--- (1)}$$

$$\text{and, } v = \frac{-y}{x^2+y^2} \quad \text{--- (2)}$$

Again, the centre of the given circle

$$z=3$$

$$\text{or, } x+jy=3.$$

$$\therefore x=3, y=0.$$

∴ Centre is at (3,0)

Also,

the radius is 2 unit.

Hence,

the eqn of the circle

$$(x-3)^2 + (y-0)^2 = 2^2$$

$$\text{or, } x^2 + y^2 - 6x + 9 = 0 \quad \text{--- (3)}$$

Again,

$$w = \frac{1}{z}$$

$$\text{or, } z = \frac{1}{w}$$

$$\text{or, } x + jy = \frac{1}{u + jv}$$

$$= \frac{u - jv}{u^2 + v^2}$$

$$= \frac{u}{u^2 + v^2} + j \left(-\frac{v}{u^2 + v^2} \right)$$

$$\therefore x = \frac{u}{u^2 + v^2} \text{ and } y = -\frac{v}{u^2 + v^2}$$

Now, putting the values of x & y in

③ we have,

$$\left(\frac{u}{u^2+v^2}\right)^2 + \left(\frac{-v}{u^2+v^2}\right)^2 - 6\left(\frac{u}{u^2+v^2}\right) + 5 = 0$$

$$\text{or, } u^2 + v^2 - 6u(u^2+v^2) + 5(u^2+v^2)^2 = 0$$

$$\text{or, } (u^2+v^2) \{ 1 - 6u + 5(u^2+v^2) \} = 0$$

$$\text{or, } 5u^2 + 5v^2 - 6u + 1 = 0$$

$$\text{or, } u^2 + v^2 - \frac{6}{5}u + \frac{1}{5} = 0$$

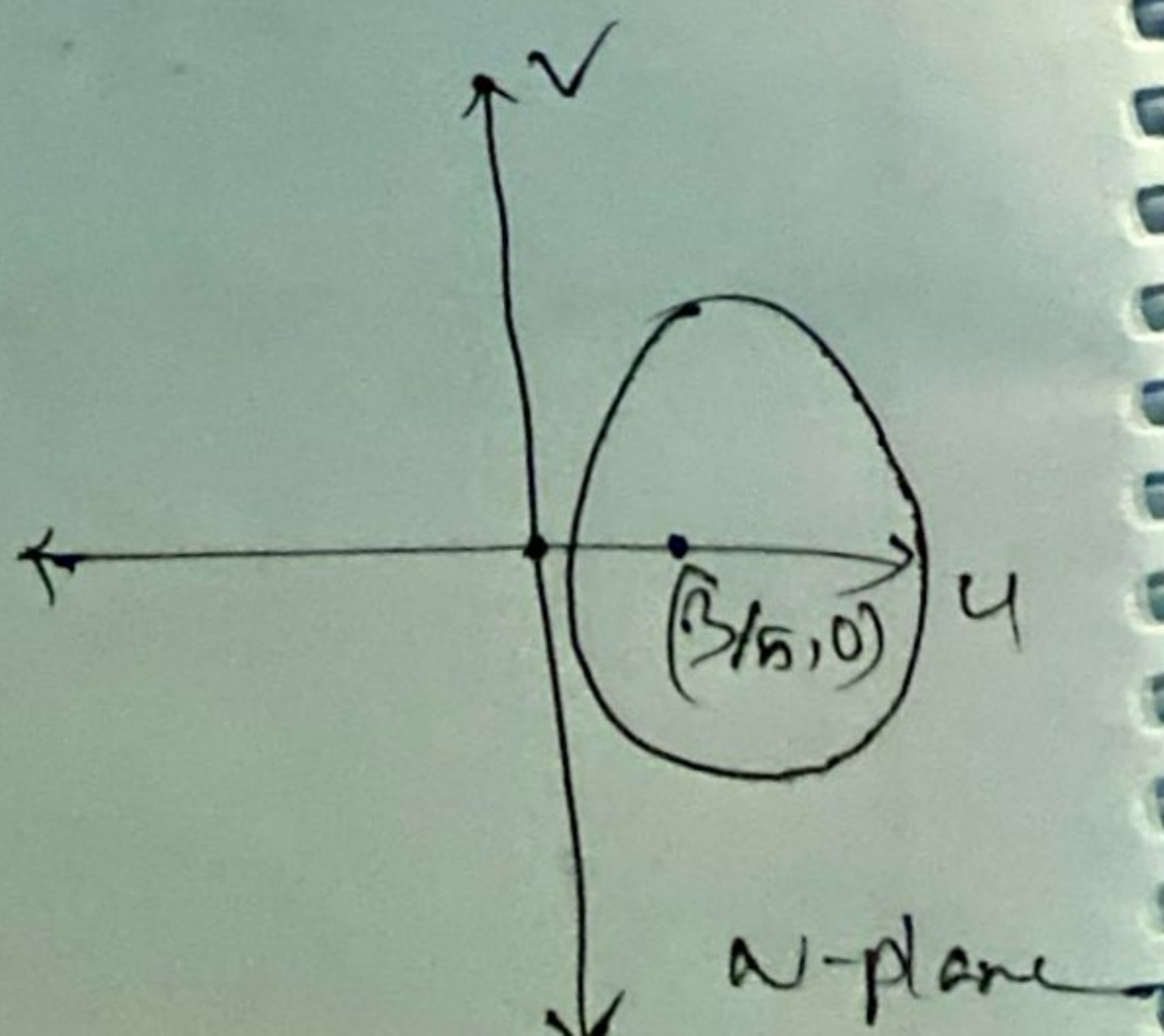
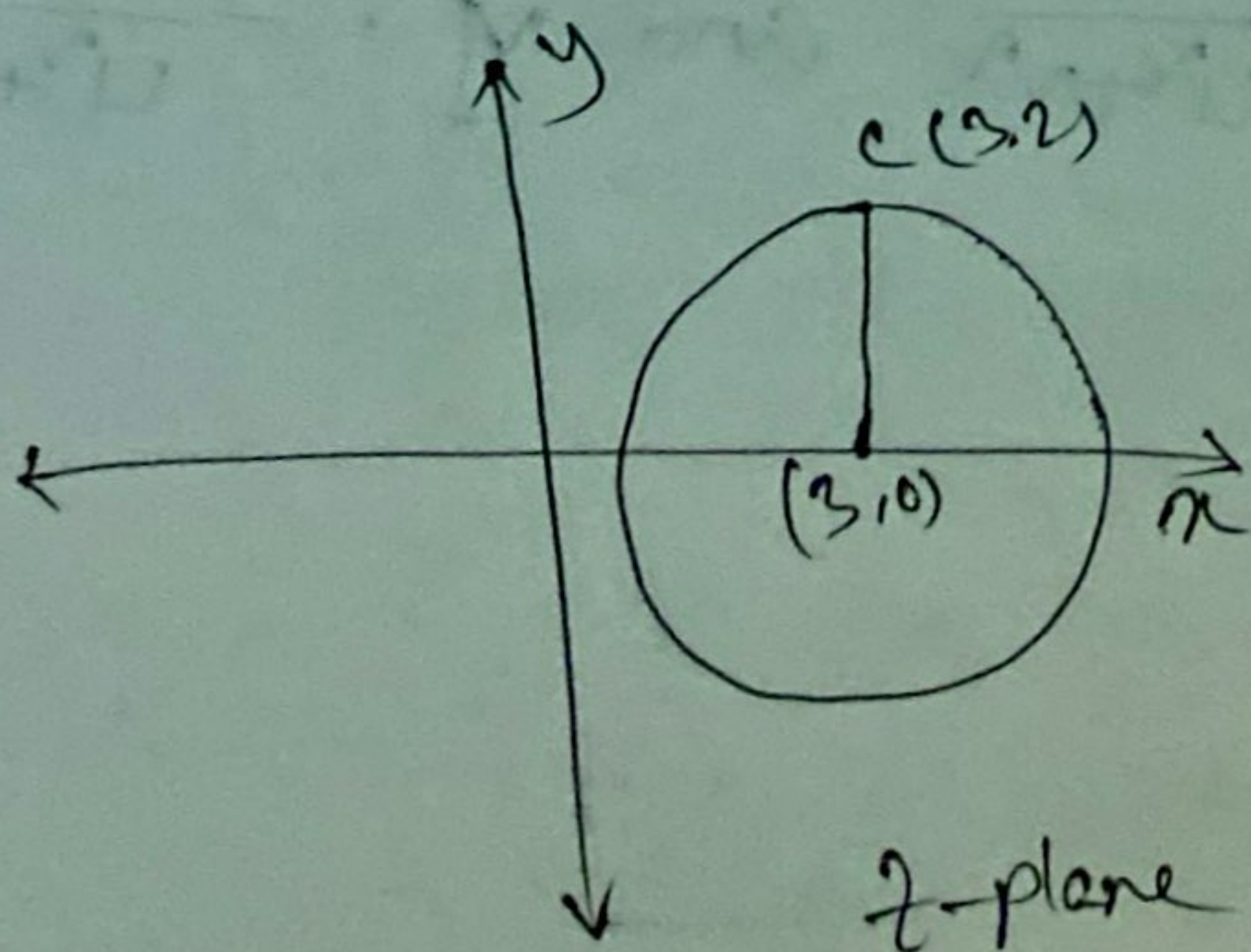
$$\text{or, } u^2 + v^2 + 2\left(-\frac{3}{5}\right)u + 2 \cdot 0 \cdot v + \frac{1}{5} = 0$$

which is the required eqⁿ of circle
whose centre is at $(3/5, 0)$ and

$$\text{radius} = \sqrt{\frac{9}{25} + 0 - \frac{1}{5}}$$

$$= \frac{2}{5}$$

Ans



Regular Function/

Analytical Function.

Cauchy - Riemann eqn

Ques:

Determine the function $w = f(z) = z^2 - 4$ is analytic or not.

Soln:

Given the function,

$$w = f(z) = z^2 - 4.$$

$$\text{or, } x^2 + j2xy - y^2 - 4 \quad [\because j^2 = -1]$$

$$\text{or, } (x^2 - y^2 - 4) + j \cdot 2xy.$$

Now, equating real and imaginary parts from both sides,

We get,

$$\begin{aligned} u &= x^2 - y^2 - 4 & \& \\ v &= 2xy. \end{aligned}$$

$$\therefore \frac{\partial u}{\partial x} = 2x$$

$$\text{Also, } \frac{\partial u}{\partial y} = -2y$$

$$\therefore \frac{\partial v}{\partial y} = 2x$$

$$\text{Also, } \frac{\partial v}{\partial x} = 2y$$

Hence,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

So the given function is analytic.

Ans

Ques:

Determine the function $w = f(z) = z^2$ is analytic or not.

Soln:

Given the function,

$$w = f(z) = z^2$$

$$\text{or, } (x + jy)(x + jy)$$

$$\text{or, } x^2 + 2j \cdot xy + (jy)^2$$

$$\text{or, } x^2 + y^2 + j \cdot 2xy$$

$$[\because j^2 = -1]$$

Now, equating real and imaginary parts from both side,

we get,

$$u = x^2 - y^2$$

$$v = 2xy$$

$$\therefore \frac{\partial u}{\partial x} = 2x$$

$$\text{Also, } \frac{\partial u}{\partial y} = 2y$$

$$\therefore \frac{\partial v}{\partial y} = 2x$$

$$\text{Also, } \frac{\partial v}{\partial x} = 2y$$

Hence,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

So the given function is analytic

Ans.

Ques:

Determine the function $w = f(z) = z^3$ is analytic or not.

Soln:

Given,

$$w = f(z) = z^3$$

$$\Rightarrow u + iv = (x + iy)^3 \quad [\because w = u + iv \text{ \& } z = x + iy]$$

$$\Rightarrow u + iv = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3$$

$$\Rightarrow u + iv = x^3 + 3x^2y \cdot i - 3xy^2 - iy^3 \quad [(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3]$$

$$\Rightarrow u + iv = x^3 - 3xy^2 + i(3x^2y - y^3) \dots (i)$$

Equating real and imaginary part.

$$u = x^3 - 3xy^2 \dots (ii)$$

$$v = 3x^2y - y^3 \dots (iii)$$

Differentiating (i) partially,

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 \quad \left| \quad \frac{\partial v}{\partial x} = 6xy \right.$$

Also $\frac{\partial u}{\partial y} = -6xy$ Also $\frac{\partial v}{\partial y} = 3x^2 - 3y^2$

Hence,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

So the given function is analytic

Ans

Ques: Test the function $v = \frac{y}{x^2+y^2}$ is harmonic or not.

Solⁿ: Given the function,

$$v = \frac{y}{x^2+y^2} \quad \text{--- (1)}$$

Now, partially diff. (1) w.r. to x two times we get,

$$\frac{\partial v}{\partial x} = y \cdot \frac{-1}{(x^2+y^2)^2} \cdot 2x$$

$$= - \frac{2xy}{(x^2+y^2)^2}$$

Also,

$$\frac{\partial^2 v}{\partial x^2} = \frac{(x^2+y^2)^2(-2y) - (-2xy) \cdot 2(x^2+y^2) \cdot 2x}{(x^2+y^2)^4}$$

$$= \frac{(x^2+y^2)(-2x^2y - 2y^3 + 8x^2y)}{(x^2+y^2)^4}$$

$$= \frac{6x^2y - 2y^3}{(x^2+y^2)^3}$$

Again,

Partially diff (1) w.r to y two times,

$$\frac{\partial v}{\partial y} = \frac{(x^2+y^2) \cdot 1 - y \cdot 2y}{(x^2+y^2)^2}$$

$$= \frac{x^2+y^2-2y^2}{(x^2+y^2)^2}$$

$$= \frac{x^2-y^2}{(x^2+y^2)^2}$$

Also,

$$\frac{\partial^2 v}{\partial y^2} = \frac{(x^2+y^2)(-2y) - (x^2-y^2) \cdot 2(2y)}{(x^2+y^2)^4}$$

$$= \frac{(x^2+y^2)(-2yx^2-2y^3-4x^2y+4y^3)}{(x^2+y^2)^4}$$

$$= \frac{2y^3-6x^2y}{(x^2+y^2)^3}$$

$$= \frac{6x^2y-2y^3}{(x^2+y^2)^3}$$

$$\text{Now, } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{6x^2y-2y^3}{(x^2+y^2)^3} + \frac{2y^3-6x^2y}{(x^2+y^2)^3}$$

As the given function satisfies the Laplace's eqⁿ, so the given function is harmonic
Ans

Ques:

Find the value of the integral

$$\int_0^{1+i} (x-y+ix^2) dz \text{ along the line from } z=0 \text{ to } z=1+i.$$

Soln:

Given that,

$$z=0$$

$$\text{or, } x+iy=0.$$

$$\text{or, } x=0, y=0.$$

Let, the point $A(0,0)$.

Also,

$$z=1+i$$

$$\text{or, } x+iy=1+i.$$

$$x=1, y=1.$$

Let the point be $B(1,1)$.

Now, the straight line joining A & B.

$$\text{is given by } \frac{x-1}{1-0} = \frac{y-1}{1-0}$$

$$\text{or, } y=x \text{ — (1)}$$

Now,

$$\int_0^{1+i} (x-y+i x^y) dz$$

$$= \int_0^1 (x-x+i+x^y) (i+1) dx \quad \left[\begin{array}{l} \text{As, } z=x+iy \\ dz = dx+iy \\ \therefore dz = (i+1)dx \end{array} \right]$$

$$= \int_0^1 (i+i^y) x^y dx$$

$$= (i+1) \left[\frac{x^{y+1}}{y+1} \right]_0^1$$

$$= \frac{1}{3} (i+1)$$

Ans

Ques: Find the residue at pole of
$$\frac{1-2z}{z(z-1)(z-2)}.$$

Soln:

let, function,

$$f(z) = \frac{1-2z}{z(z-1)(z-2)}.$$

Now, for pole,

$$z(z-1)(z-2) = 0.$$

$$\therefore z = 0, 1, 2.$$

Now, residue of $f(z)$ at $z=0$,

$$\lim_{z \rightarrow 0} (z-0) f(z).$$

$$= \lim_{z \rightarrow 0} z \cdot \frac{1-2z}{z(z-1)(z-2)}$$

$$= \frac{1-2 \cdot 0}{(0-1)(0-2)}$$

$$= \frac{1}{2}$$

Ans

Now,

Residue of $f(z)$ at $z=1$,

$$\lim_{z \rightarrow 1} (z-1) f(z).$$

$$= \lim_{z \rightarrow 1} (z-1) \left(\frac{1-2z}{z(z-1)(z-2)} \right)$$

$$= \frac{1-2 \cdot 1}{1 \cdot (1-2)}$$

$z=1$

And,

Residue of $f(z)$ at $z=2$,

$$\lim_{z \rightarrow 2} (z-2) f(z)$$

$$= \lim_{z \rightarrow 2} (z-2) \frac{1-2z}{z(z-1)(z-2)}$$

$$= \frac{1-2 \cdot 2}{2 \cdot (2-1)(2-2)}$$

$= 0$.

Ans